

Video Highlight Segment Recognition And Automatic Editing Method Based on Spatio Temporal Convolution Network and Multimodal Emotion Alignment

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Abstract—The current AI clips are deeply involved in shallow feature matching, resulting in a physical stack lacking narrative tension. In order to bridge the audio-visual semantic gap, this paper proposes an automatic editing method combining spatio-temporal convolution network (st-gcn) and multimodal emotion alignment. Firstly, the model uses st-gcn to carry out topological modeling of scene scheduling, and generates "action tension curve" to provide objective physical editing cut points; At the same time, the heterogeneous characteristics of chorus, painting and writing are forced to output the "emotional resonance curve" to deeply perceive the narrative fluctuations. The case shows that the highlight extraction and art combination driven by the intersection of two streams significantly surpass the traditional algorithm in the objective hit rate and subjective double-blind aesthetic test, and the rhythm and empathy of the film are highly close to human experts, successfully promoting the leap of AI from a mechanical splicing tool to an intelligent audio-visual narrative subject.

Keywords—*Spatio-Temporal Graph Convolutional Network (ST-GCN); Multimodal Emotion Alignment; Video Highlight Segment Recognition; Automatic Video Editing*

I. INTRODUCTION

With the explosive growth of massive digital media content, the contradiction between the huge demand of video production and traditional manual editing has become increasingly prominent. To meet this challenge, a large number of "one click editing" systems have emerged in the market. However, the current intelligent editing often falls into the technical dilemma of "focusing on efficiency and ignoring emotion". Most tools only focus on the pattern matching of shallow features, such as mechanical segmentation based on the threshold of image brightness, face area or audio loudness. This fragmented and lack of emotional insight leads to the fact that the generated video is often just a physical stack of materials, which seriously lacks the "narrative climax" and "internal rhythm" of the audio-visual language of film and television, and is difficult to really impress the audience.

To break through the limitations of current AI editing tools, the first prerequisite is to clarify the essence of film and television highlights. The classic film and television editing theory points out that highlight is by no means a simple visual explosion or high decibel cry. It originates from the deep superposition of two core dimensions: one is the action conflict in physical space, that is, the objective momentum of "who did what to whom"; The second is the emotional resonance in the psychological space, that is, "the joys and sorrows transmitted by the character state or the sound and painting atmosphere". Unfortunately, the existing single algorithm can only understand the motion of pixels or hear the fluctuation of waveform, which is difficult to cross the semantic gap from the underlying data to the high-order drama tension.

Based on the above pain points, this paper proposes a collaborative method for film and television highlight segment recognition and automatic editing, aiming to reconstruct the film and television aesthetic rules with computable logic. In the field of computational vision, spatiotemporal graph convolution network (st-gcn) is a cutting-edge network dedicated to understanding the "mesh structure moving with time". In this study, we abstract the skeleton joints and key props in the picture as "nodes", connect the physical distance between them as "edges", and then construct a dynamic spatio-temporal map. Over time, the distortion, deformation and interaction frequency of this picture can accurately and objectively calculate the "intensity of action" and "spatial scheduling relationship" in the picture. At this time, st-gcn can capture the objective physical motion and is responsible for drawing the "action tension curve" throughout the film. On the other hand, film is a comprehensive art composed of pictures, sounds and lines. In this study, multimodal emotion alignment technology is introduced to let AI learn to "master". This technology breaks the limitation of independent processing of vision or hearing in the past, and forces vision, hearing and text to be pulled into the same high-dimensional space for comparison and synchronization. This ensures that AI can not only understand the sadness in the lines, but also see the

loneliness in the picture. Therefore, multimodal alignment technology can perceive subjective emotions and give the system the ability to perceive "dramatic tension".

To sum up, through the mathematical intersection of "objective physical tension" and "subjective emotional resonance", this paper not only realizes the accurate extraction of film and television highlights, but also lays a theoretical foundation for the subsequent automatic assembly based on aesthetic rules, and promotes the paradigm transition of AI editing from mechanized production to emotional creation.

II. LITERATURE REVIEW

The research context of intelligent video editing has experienced the evolution from low-level feature extraction to high-level semantic understanding. In terms of video summarization and highlight extraction, early studies mostly relied on random frame extraction. With the development of deep learning, Wang Ping and other scholars pointed out that relying on computer vision and natural language processing technology, machines can automatically filter content according to structural features [1]. The research of men Fei and others further found that although the automatic processing based on single source feature significantly improved the production efficiency in short video quick cutting, it had significant limitations [2]. Chen Jie and other scholars believe that the isolated signal of vision or audio can easily make the model fall into the trap of "local extremum", such as misjudging the abrupt noise as the narrative climax, but unable to perceive the text tension under the calm picture, resulting in the lack of coherent story line and narrative arc in the extracted summary [3].

In order to bridge the semantic gap between the underlying physical signal and audio-visual language, "computational filmology" has gradually become the research focus. Some scholars have devoted themselves to translating the classic theories of director and editing, such as scene matching and dynamic cohesion, into computable mathematical rules. At the level of video action understanding, Duan Xinran and other scholars found that the introduction of graph networks can effectively analyze the topological interaction between characters and scenes in long videos [4]. However, industry experts generally believe that most of the current video analysis based on graph network still stays at the macro level of action classification, and fails to deeply bind it with the aesthetic principle of "moving and touching" in film and television editing. When looking for the best cutting point, the machine is still limited by the coarse inter frame difference, and cannot accurately capture the extreme point of limb force as the anchor point of physical tension [5].

In the direction of audio-visual rhythm and emotion driven editing, in recent years, a large number of algorithms have tried to simulate the human trampling editing art [6]. Zhangkang, songshien and other scholars achieved

mechanical alignment between video hard cutting and music drums by extracting BPM beats of background music [7]. Although some advanced studies began to explore the match between emotional tone and picture, scholars such as Gao Meng pointed out sharply that the current intelligent editing system generally presents the characteristics of "focusing on efficiency and ignoring emotion". They believe that AI assisted editing at this stage focuses too much on templating and mass production, and is still essentially a tool lacking empathy [8]. This shallow label mapping mechanism failed to really break the modal barrier between vision, such as warm and cold colors, hearing, such as melodic fluctuations and text, resulting in AI's serious lack of cognition of the deep emotional context of film and television [9].

To sum up, the existing technology is either limited by the surface features of single mode, or trapped in mechanical splicing lacking emotional depth, and has never touched the artistic essence of "superposition of physical conflict and psychological emotional resonance" of film and television highlights. Based on the above academic consensus and technical blind spot, this paper proposes a new paradigm integrating spatio-temporal convolution network and multimodal emotion alignment, in order to achieve a fundamental breakthrough in the field of automatic editing.

III. RESEARCH METHODS

Aiming at the technical bottleneck of the current intelligent editing system, which emphasizes efficiency and ignores emotion, this paper overcomes the limitations of traditional matching based on surface pixels or single source features, and proposes a dual stream computing framework that integrates physical momentum and psychological emotion.

A. Overall Process of the Model: Workflow Design of dual Flow Feature Collaboration

The macro workflow of this study consists of three closed-loop modules: feature extraction, highlight evaluation and rule driven. The overall process is shown in Figure 1. After the video sequence is input, the system starts two analysis links in parallel. The space-time convolution network (st-gcn) module focuses on the human skeleton trajectory and scene scheduling in the picture, and extracts the objective physical tension characteristics; The multimodal emotion alignment module deeply analyzes the heterogeneous data of image, sound and text, and extracts the subjective emotional resonance features. Then, the system weighted the above physical and psychological features through the cross modal attention mechanism, and calculated the continuous highlight score of the video frame. Finally, according to the extreme value distribution of highlight score, the system triggers the built-in film and television aesthetic editing rules, such as action cohesion and emotional montage, to automatically complete the smooth assembly and film output of discrete shots.

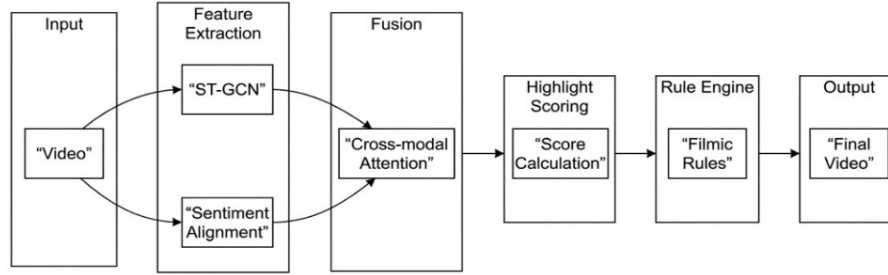


Fig 1. Overall flow chart of the model

B. Surface Scheduling And Physical Tension Calculation Based On Space-Time Graph Network

In film and Television Aesthetics, scene scheduling is the core of constructing visual tension and narrative power. In order to realize the computability of complex scheduling, st-gen is introduced to model the physical motion in the picture. Firstly, the model automatically identifies the spatial position and bone key points of the main characters in the picture based on the pose estimation algorithm, and maps them to the graph nodes in the spatial dimension. Then, along the time line of the video frame, the system tracks these key points with high precision, and constructs a spatio-temporal topology representing the continuous trajectory of the character's actions. On this basis, the st-gcn network quantifies the physical conflict through the deformation intensity of the graph structure. For example, when the picture shows people sitting and talking, the spatio-temporal topology remains stable, and the system outputs a low tension score; However, in the event of action intensive segments such as chasing or physical conflict, the graph structure will produce high-frequency nonlinear distortion and spatio-temporal interaction. By calculating the gradient and amplitude of topological deformation, the model converts it into quantitative tension value, and finally generates a physical action tension curve through the time axis. The curve accurately anchored the extreme point of the action force, and established an objective kinematic basis for the selection of the cutting point of the clip.

C. Audio Visual Emotion Extraction Based on Multimodal Semantic Alignment

Film and television art is a comprehensive medium with a high degree of synergy among picture, sound and lines.

Relying solely on the single source characteristics of acoustic mutation or visual saliency can easily lead to the misjudgment of machine narration. Therefore, based on the multimodal emotion alignment theory, this paper constructs a calculation model for perceiving the internal emotional rhythm of the film. The workflow of the model is shown in Figure 2. In the feature perception phase, the system deploys three parallel deep learning networks to scan the heterogeneous modes. The visual branch analyzes the warm and cold tone of the picture color and the micro expression of the actors' faces; The audio branch uses acoustic model to extract the rhythm and mood fluctuation of background music; Text branches use natural language processing technology to quantify the emotional polarity of subtitle semantics [10]. In order to solve the semantic ambiguity between modes, this study innovatively introduces the cross modal alignment algorithm. The mechanism strictly judges whether the three underlying signals point to the same high-order emotion in the same time window through the joint embedding space. For example, only when the visual features of actors' tears and the acoustic features of sad strings highly coincide with the text semantics of farewell lines, the model will label them as emotional highlight nodes. This deep semantic alignment effectively filters the modal noise interference caused by monotonic brake noise. Finally, the module outputs an emotional resonance curve representing the fluctuation of the film's psychological state, which complements the depth of the physical tension curve, and constitutes the core core of driving the intelligent editing system to achieve empathy expression.

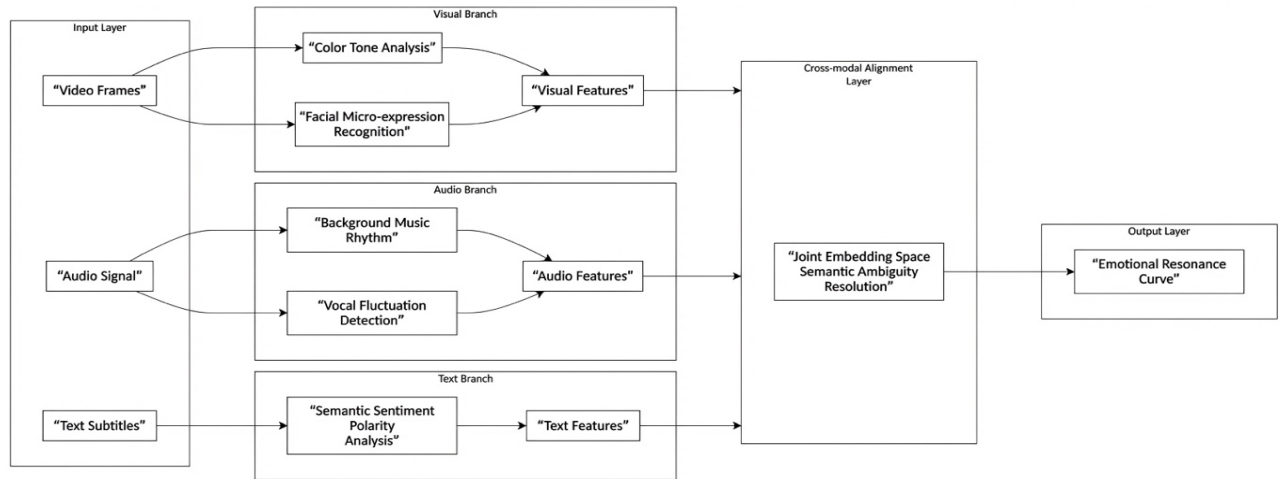


Fig 2. Flow chart of emotion extraction based on multimodal semantic alignment

D. recognition Mechanism of Video Highlights

After obtaining the physical action tension curve output by the spatio-temporal map network and the emotional resonance curve output by the multimodal module, this study constructed a highlight segment recognition mechanism based on the intersection of two streams. Traditional video abstracts often only rely on a single extreme value for threshold truncation, which is easy to extract physical fragments lacking context. In order to overcome this limitation, this module introduces an adaptive weighted fusion algorithm, which superimposes the "action intensive point" representing the objective scheduling and the "emotional peak" representing the subjective tension on a unified time axis. By setting a dynamic threshold, the system accurately peels off the golden audio-visual interval where physical conflict and psychological resonance overlap. These intersections not only contain external high-energy visual stimuli such as chase and fierce physical conflict, but also carry the emotional explosion of the plot core [11]. After time-series clustering and boundary smoothing, the algorithm finally finds the resonance interval with the highest score in the global dimension of the whole film, screens and outputs a pool of film and television highlight candidates with high semantic density to provide high-quality material carriers for subsequent artistic assembly. The whole process identification mechanism is shown in the following Table1.

TABLE I. VIDEO HIGHLIGHT SEGMENT RECOGNITION FLOW TABLE

Module	Input	Core Function	Output
Input Layer	Physical Tension Curve Emotional Resonance Curve	Dual-curve parallel input, providing data foundation for subsequent fusion	Two feature curves
Adaptive Weighted Fusion	Physical Tension Curve Emotional Resonance	Weight adaptation Temporal superposition	Fused dual-stream features

	Curve	alignment	
Dynamic Thresholding	Fused dual-stream features	Action peak detection Emotional peak detection Overlap determination	Overlapping candidate intervals
Temporal Refinement	Overlapping candidate intervals	Temporal clustering Boundary smoothing	Refined candidate intervals
Global Scoring	Refined candidate intervals	Highest resonance interval selection	High-scoring candidate intervals
Output Layer	High-scoring candidate intervals	Highlight segment pool generation	Semantically dense highlight candidate pool

E. Automatic Editing And Assembling Algorithm Based on Film Aesthetics

As the core of this computing framework, the automatic editing and group connection algorithm aims to endow the machine Hollywood with industrial editing thinking and get rid of the defects of rigid splicing of traditional AI tools. For the discrete highlight candidate segments, the system transforms the classic action cohesion criteria and emotional montage theory into rigorous mathematical optimization objectives.

First, in the accurate selection of clip points, the system directly calls the action tension feature extracted by st-gen to find the action vertices in the frame sequence. The algorithm uses the extreme frame of the action derivative, such as the moment when the fist is wielded to the highest kinetic energy, as the entry and exit anchor of the lens, and uses the visual persistence effect to cover the montage gap. This "move to move" strategy based on dynamic calculation ensures the flow of adjacent shots in the dynamic connection, and avoids the visual separation caused by jump cut. Secondly, at the level of audio-visual rhythm matching and narrative reorganization, the system dynamically adjusts the

editing rate according to the multimodal emotion score and audio beat [12]. When the multimodal feature indicates that the current scene is in a climax mood or a paragraph with dense drums, the algorithm automatically triggers the "quick cut" mode, which greatly reduces the retention time of a single shot and pushes up the audio-visual pressure with high-frequency chopping; On the contrary, in the soft emotion or soothing segment, the system switches to the "long shot" mode to retain the integrity of the action and provide audio-visual breathing space for emotional precipitation. Finally, the algorithm relies on the macro emotional envelope of the whole film, carries out global serialization and reorganization of the candidate fragments, and automatically constructs a film structure that conforms to the micro narrative arc of "starting, inheriting, transferring and combining". This mechanism truly realizes the deep mutual consistency between the underlying computing logic and the top-level film aesthetics. The overall workflow is shown in Figure 3.

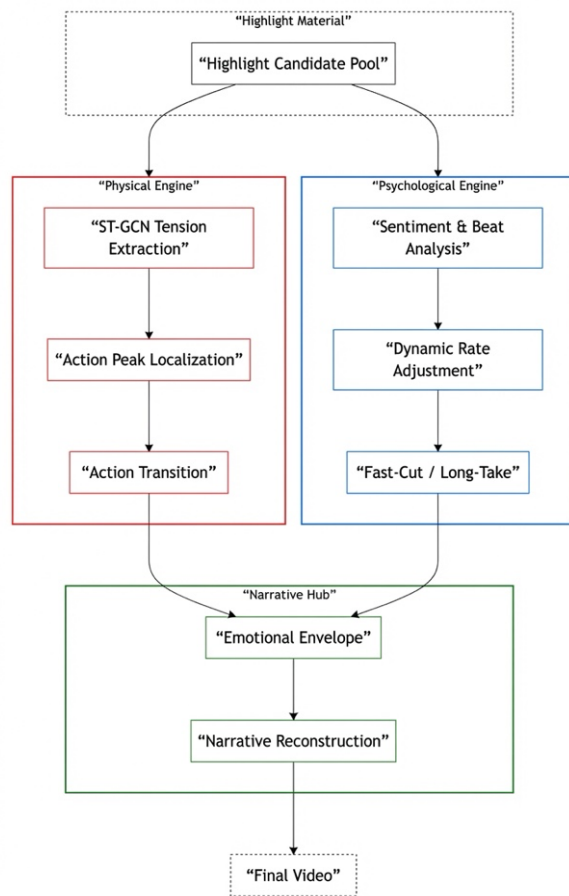


Fig 3. Structure diagram of automatic editing and assembly algorithm based on film and Television Aesthetics

IV. CASE ANALYSIS

A. Comparison and Qualitative Analysis of Automatic Editing Effect

In order to verify the automatic editing effect of this model in different types of films, this section selects classic action films and romantic films for 1-minute highlight film test, and in the paper, the "action tension curve" and "emotional resonance curve" output by the system are synchronously mapped through the graph and text mirror table. In the action test, we extracted the famous "sandstorm chase" in crazy Max 4: the road to fury. Traditional AI clips often rely on the audio loudness threshold, which is very easy to cut off when the action is not completed, resulting in abrupt drop and jump cut of the picture, as shown in Figure 4. In this model, the "action tension curve" output by st-gcn accurately generates a crest when a character waves a weapon and collides with a vehicle. Taking the lens of nakes throwing a spear as an example, AI accurately locked the extreme frame of the action when the arm was swinging to the highest point and the spear was about to be released, and set it as the cut-out of the current lens; Then, seamlessly connect the cut in of the enemy combat vehicle in the next shot, as shown in Figure 5. This splicing, which is strictly driven by the extreme points of the action curve, ensures the flow of the two discrete shots in the physical momentum, and shows the spatial scheduling level comparable to that of the senior human editors in the film.



Fig 4. Case of traditional AI editing





Fig 5. AI clip based on spatio-temporal graph convolution network

In the emotional test, we selected the classic "fantasy montage" paragraph at the end of the city of love. In this 1-minute highlight video, the "emotional resonance curve" in the first 40 seconds presents a high-frequency oscillation with the rapid conversion of screen color and jazz drums. At this time, AI automatically shortens the retention time of each shot, and reorganizes the "missed perfect life" of the male and female protagonists with fast cutting. However, the most academically valuable decision appeared at the end of the short film: when the two people looked at each other and smiled at each other in the bar, the physical movement in the picture was still, and the "action tension curve" suddenly dropped to a very low value; However, through multimodal perception, the actor's Micro expression and background tone without dialogue push the "emotional resonance curve" to the climax of the film. In the face of this contrast, AI does not cut off the static picture like the conventional summary algorithm, but automatically triggers the narrative protection mechanism, and retains the staring long shot of up to 12 seconds as the final frame.

The real case above shows that the hyperbola output by this algorithm can not only accurately anchor the cut-in and cut-out points of the lens, but also enable the machine to truly grasp the relaxed "breath" in film and television editing, and realize the leap from mechanical splicing to aesthetic creation.

B. Highlight Hit Rate and Audio Visual Blind Test Based on Expert Consensus

In order to scientifically quantify the editing level, several senior editors were invited to dot the sample film frame by frame in the experiment, so as to build the human aesthetic standard. Statistics show that, compared with the traditional fast cutting tools which only rely on the volume or brightness mutation, the model in this paper effectively filters the meaningless sudden impact sound, avoids misjudging the local extreme value as the narrative climax, and makes the core highlight hit rate achieve a significant jump.

At the subjective perception level, this study carried out a rigorous double-blind test of film and Television Aesthetics. The evaluation matrix composed of professional scholars and the public audience scores the film without knowing the identity of the editor. The results show that thanks to the precise implementation of st-gcn's physical laws of action cohesion and the deep fitting of multimodal emotions, the film generated by this model completely gets rid of the

machine sense of raw material. Its comprehensive score in the dimensions of "smoothness of lens connection" and "emotional appeal" is not only far beyond the traditional AI, but also highly close to the professional intuition of senior human editors, which truly verifies the leap from AI to intelligent audio-visual narrative subject.

V. CONCLUSION

Aiming at the pain point of the current intelligent editing system, which focuses on efficiency and ignores emotion, this paper proposes and verifies a dual stream automatic editing framework integrating spatiotemporal graph convolution network (st-gcn) and multimodal emotion alignment. This study is the first to deconstruct film and Television Aesthetics as the mathematical intersection of "physical tension" and "emotional resonance". Among them, st-gcn accurately anchors the action extreme value through spatio-temporal topology deformation, and realizes the industrial level "dynamic connection" spatial scheduling calculation; The multimodal alignment module breaks through the characteristic barriers of sound, painting and writing, and endows the system with the empathic ability to perceive the deep dramatic tension and narrative rhythm. The multi-dimensional case analysis and double-blind aesthetic test confirm that this model effectively resolves the semantic ambiguity and local extremum trap of traditional single source features. The automatically generated highlight film not only achieved a significant jump in the extraction hit rate of core clips, but also highly approached the professional intuition of senior human editors in the two core dimensions of smoothness of lens connection and emotional appeal.

To sum up, this study successfully crossed the semantic gap between the low-level pixels and the high-level audio-visual language, and reconstructed the editing art with rigorous computational logic. This marks that AI has officially bid farewell to the rigid physical splicing in the field of video editing. In the future, we will continue to explore the global generation system of long video macro narrative arc, and further broaden the academic boundary of computational filmology.

REFERENCES

- [1] Wang Ping, fanyonghong Research on the application of artificial intelligence technology in short video automatic editing [j]. network audio visual, 2025, (09):81-83.
- [2] Door flies Application of intelligent audio and video editing and automatic generation technology [j]. home theater technology, 2025, (12):16-19.
- [3] Chen Jie Research on the application of AI technology in micro video editing in the context of intelligent editing [j]. reporter's cradle, 2025, (11):141-143.
- [4] Duan Xinran, Wang Mei, Han Tianli, et al Perception and analysis of teaching process based on video understanding [j]. computer science, 2024,51 (10): 56-66
- [5] Chengbaokai, Dai Chaoyue, sunmengyan Research on the application of artificial intelligence in film and television interpretation and automatic editing [j]. information recording materials, 2025,26 (9): 237-241245

- [6] Zeng Boyi, houjianzhong The influence of computer vision editing technology in film production [j]. home theater technology, 2025364 (14): 47-50
- [7] Zhang Kang, song Shien, Wang Xinying, et al Research and application of short video automatic production based on artificial intelligence [j]. radio and television technology, 2024,51 (12): 31-35
- [8] Gao Meng, hanjinshan, Zeng Zhigang Evolution path and Paradigm Innovation of intelligent editing [j]. Journal of Beijing Film Academy, 2025, (12):93-103.
- [9] Meng Xiang, Li Sha Research on the teaching practice of video post editing driven by rhythmic thinking [j]. art and technology, 2024,37 (22): 176-181
- [10] Zeng Boyi, Zheng Bing Multimodal emotion recognition and content intelligence enhancement method in AI driven video editing [j]. Changjiang information and communication, 2025,38 (10): 117-119122
- [11] Lizijian Research and application of video emotion recognition based on multimodal fusion [d]. Chongqing: Chongqing Normal University, 2024
- [12] Wanglingyun, Longhu Exploration on Ideological and political path of film and television editing techniques and Application Course under multimodal theory [j]. communication and copyright, 2025 (1): 100-102.